



speaks to a growing
number of

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EFY Enterprises Pvt. Ltd.
D-57/1, Okhla Industrial Area, Phase I, New Delhi 110 020
Ph: 011-26510601-43; Fax: 011-26517563, 26612312
e-mail: info@efyindia.com website: www.efyindia.com

demoralising effect on the efforts being put in by the Institute:

"Scientists at the IISc, Bangalore, have been working on biomass gasification for more than 20 years. They have developed fully commercial technology for producing heat or power from wastes. Though the technology is being implemented in several industries, certain upstream R&D is still required to modify the technology for proper utilisation of refuse-derived fuel.

"We were planning to try this technology through a pilot plant at a small town called Chitradurga where the estimated daily municipal solid waste production is 44 tonnes (15 tonnes of combustible material). The plant involves processes like segregation, drying and pelletisation of municipal waste before gasification with about 220 kW of power production. Of this, about 50 kW was estimated for in-house consumption. Under the plan, the plant was to be set up on a landfill site spread over an area of 14 hectares on the outskirts of the town. The project had also received approval from Chitradurga municipality.

"However, following a hullabaloo by well-intentioned environmentalists, the whole matter is now in a limbo."

Import of foreign technologies for waste management. We should wisely evaluate all pros and cons before adopting any imported technology. We must draw lesson from the costly blunder that was made in 1980's while installing a Danish-supplied incinerator at Timarpur in Delhi at a cost of Rs 410 million that ran for only six days because of high inerts and ash in the waste supplied. India lost the case at Hague because 'suitable waste' for be provided was not defined in the agreement. Current interest and caretaking costs have totalled a staggering Rs 2 billion or more. Even worse, this valuable waste-disposal site has been lost for decades to the city as the plant is

neither being sold as scrap, nor are there any takers to take it over for operation.

Assessing viability of the imported plasma gasification technology. For assessing the financial viability of the projects, especially those involving foreign firms like Startech, the operating cost, net cost of operation and capital cost should be taken into account.

Operating cost. The electrical power requirement for conversion of one tonne of municipal solid waste into vitrified solids and metals, hydrogen and carbon monoxide gas averages around 670 units (kWh). At Rs 3 per unit, the cost of conversion works out to be Rs 2000 per tonne. Although one may be able to reduce the running cost by, say, 75 per cent, after selling the byproducts, the net cost of the plant would still be Rs 750 per tonne.

Net cost of operation. As an example, let us consider Delhi (not entire NCR) with its current population of over 15 million. The average garbage generation may be assumed to be 0.7 kg per person per day (against Americans' average of about 2 kg per person per day). Accordingly, Delhi would be generating around 10,500 metric tonnes of garbage per day. The net operational cost would thus be Rs 7.875 million per day (at Rs 750 per tonne) or Rs 2870 million per annum. This is in addition to garbage collection and transportation and other infrastructure costs.

Capital cost. A Startech plasma converter that could handle 2000 tonnes of waste daily costs roughly \$250 million. Delhi would require eight such huge plants at a whopping cost of \$1000 million (roughly Rs 40 billion).

India, being a labour-intensive country, should employ matching technologies by segregating the various types of wastes and treating each one using indigenously-developed appropriate technologies. ●

Concluded